

Endless AI: A Decentralized Intelligent Creator-Centric Ecosystem

Endless AI

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Abstract

The convergence of artificial intelligence (AI) and Web3 promises unprecedented innovation but presents significant challenges concerning ethical deployment, intellectual property (IP) protection, privacy, and fair compensation within a decentralized framework. Existing centralized models struggle with data sovereignty and intermediary dependence. We introduce Endless AI, a decentralized Intelligent Creative Ecosystem designed to revolutionize the intersection of AI, Web3 infrastructure, and creator economies. Endless AI empowers creators by enabling on-chain verification, monetization, and governance of their IP using blockchain-authenticated NFTs and tokenized incentives within the EDS ecosystem. Leveraging Agentic AI, Model Context Protocol, cross-chain AI agents, and decentralized storage, the platform facilitates autonomous asset management and royalty enforcement. Privacy is preserved through advanced techniques like zero-knowledge proofs and KZG polynomial commitments. Built upon the Endless Web3 Genesis Cloud architecture, the system ensures payment assurances and democratic governance, tackling the core challenges of deploying AI responsibly and effectively in the decentralized future. Endless AI provides the framework for a more equitable, secure, and innovative digital ecosystem for creators and AI alike.

1. INTRODUCTION

The Web3 landscape has witnessed significant growth yet many platforms have become dominated by financial products and speculation, often neglecting the foundational principles of empowering users, protecting privacy, and ensuring security. True Web3 innovation should focus on enhancing user experience, safeguarding privacy, and enabling user control over data, while simultaneously unlocking new economic opportunities. Endless Web3 Genesis Cloud is designed to embody this core vision, providing a flexible, privacy-preserving, and data-secure decentralized development platform. It lowers technical barriers for developers, facilitates Web2 application migration, and supports diverse scenarios beyond finance, delivering real-world value to the ecosystem. By providing comprehensive privacy, data sovereignty, and a collaborative

value-creation environment, Endless aims to return Web3 to its original promise of genuine user empowerment and value co-creation, stimulating user-driven value creation and propelling the digital economy forward. The integration of artificial intelligence, particularly generative AI and agentic AI into this decentralized framework is the next frontier poised to revolutionize content creation, intellectual property management, and asset monetization within a truly decentralized creative ecosystem.

The digital age has ushered in an unprecedented era of creativity, driven by the rapid advancement of artificial intelligence and generative content tools. From AI-generated art and music to automated code generation and virtual world-building, AI is transforming how content is created, distributed, and monetized. However, this innovation comes with critical challenges, intellectual property (IP) theft, centralized platform control, and opaque monetization models leave creators vulnerable to exploitation. Traditional Web2 platforms, such as OpenAI, Midjourney, and Adobe Firefly retain ownership over user-generated data, dictate revenue splits, and offer no verifiable proof of authorship. Meanwhile, the rise of deepfakes, plagiarism, and unauthorized AI training threatens the integrity of creative works. The Endless AI is a decentralized, blockchain-powered framework that redefines the relationship between AI, creators, and digital ownership. Built atop the Endless Web3 Genesis Cloud¹, Endless AI integrates Agentic AI, decentralized storage, and tokenized incentives to establish a self-sovereign creative ecosystem. Unlike centralized AI platforms, Endless AI ensures that creators retain full control over their works, from initial creation to monetization, while leveraging blockchain to verify authenticity, enforce royalties, and reward participation.

1.1. Motivation

The rapid rise of artificial intelligence generated content (AIGC) has exposed critical flaws within the existing digital ecosystem, creating a legal and economic gray area that disadvantages creators and stifles innovation. The core motivation behind Endless AI stems from the urgent need to address these systemic failures,

- **LACK OF INTELLECTUAL PROPERTY PROTECTION:** AI-generated works often fall outside traditional copyright protections. This leaves creators vulnerable with their novel digital paintings, AI-trained musical compositions, or other creations unprotected against plagiarism and unauthorized use. There is ambiguity concerning ownership – who truly owns the output when AI is involved? Existing frameworks fail to provide clear answers or recourse.

¹<https://endless.link>

- **EXPLOITATIVE CENTRALIZED PLATFORMS:** The dominant Web2 platforms (such as Midjourney, OpenAI, etc.) operate on centralized models. They control vast amounts of user data often monetizing it without fairly compensating the creators whose inputs and interactions generate value. Also, creators face restrictions on data portability locking them into specific platforms and hindering their ability to leverage their work elsewhere. This reliance on intermediaries leads to opaque algorithms and arbitrary policy changes that creators have no control over.
- **INADEQUATE AND FRAGMENTED MONETIZATION:** While technologies such as NFTs have emerged for digital ownership, traditional NFTs often represent static assets. They lack the dynamic royalty mechanisms needed for AI-generated works which can evolve, be remixed, or used in various contexts over time. This fragmentation makes it difficult for creators to capture the ongoing value of their AI creations effectively.
- **EROSION OF TRUST AND STIFLED INNOVATION:** The combination of unclear ownership, unfair compensation, platform lock-in, and lack of control forces creators to operate within a broken system reliant on middlemen. This environment fundamentally erodes trust between creators and platforms and ultimately stifles the true innovative potential of AI in the creative fields.

Endless AI emerges as a direct response to this dysfunctional status quo. It is motivated by the vision of a creator-centric future built on the principles of decentralization, transparency, and fair compensation. By strategically integrating AI's generative power with Web3's ownership principles (blockchain and tokenomics), Endless AI aims to:

1. **Empower Creators with True Ownership:** Provide tools for one-click IP confirmation, minting AI works as verifiable, royalty-bearing NFTs with immutable provenance recorded on-chain.
2. **Automate and Decentralize Management:** Deploy autonomous AI agents governed by smart contracts to manage licensing, detect plagiarism, and distribute payments automatically and transparently, removing the need for costly intermediaries.
3. **Establish Fair Incentive Structures:** Implement a tokenized ecosystem (EDS) that rewards creators, data providers, AI model trainers, and validators for their contributions, ensuring value flows directly to those who build the ecosystem.
4. **Enable Advanced & Privacy-Preserving AI:** Utilize innovations like the Model Context Protocol for context-aware generation and Zero-Knowledge Proofs (ZKPs) for privacy-

preserving AI training, fostering trust and sophistication.

5. **Foster an Open, Composable Ecosystem:** Operate as an open protocol that integrates with existing creative tools and enables entirely new, decentralized business models (like AI-NFT licensing, Data DAOs, and AI Curation Markets).

The motivation for Endless AI is to initiate a paradigm shift away from exploitative, centralized control towards a self-sustaining, decentralized creative economy. It seeks to build a future where algorithms are accountable, IP is protected, innovation is rewarded, and creators can finally thrive without compromise, reclaiming control and value in the age of AI.

2. ENDLESS AI AND ITS KEY MECHANISMS

Endless AI is positioned as a decentralized framework that seamlessly integrates artificial intelligence with the Endless Web3 infrastructure and blockchain technology. Its primary goal is to enable the creation of self-improving, incentivized AI applications that operate within a secure and privacy-preserving environment. The Endless Web3 infrastructure provides robust support for AI through several key mechanisms,

- **On-Chain Learning:** Endless AI models can directly access and process data stored on the blockchain, including transaction histories, smart contract interactions, and potentially, decentralized data streams from various DApps. This on-chain access allows AI models to learn and refine their decision-making processes based on real-time, transparent, and immutable data, which is critical for applications like decentralized finance (DeFi) analysis or on-chain governance.
- **Incentivization via Tokenomics:** A core aspect of Endless AI is its token-based incentive system, leveraging the native EDS tokens. AI agents, developers, and infrastructure providers are incentivized to contribute to the ecosystem. This includes earning EDS tokens for tasks such as training AI models, validating data used by AI, or deploying and maintaining AI-driven decentralized applications (dApps). This creates a self-sustaining economic model where contributions to the AI ecosystem are directly rewarded.
- **Decentralized Data Sovereignty:** Ensuring user privacy and data control is paramount in the Web3 era. Endless AI operates on encrypted, user-owned data. This data is stored on Endless's decentralized storage network utilizing technologies like IPFS and a key-value storage system for efficiency and security. This decentralized approach combined with privacy-preserving mechanisms like ZKPs and dynamic end-to-end encryption, ensures that

AI computations and applications respect user data sovereignty and comply with privacy requirements.

The key mechanisms powering Endless AI incentivization are,

1. **Staking Rewards:** AI service providers and those contributing computational resources stake EDS tokens to participate in the network. By staking, they commit to providing reliable and accurate AI services (e.g., providing data for AI training, running inference models). They earn staking rewards in EDS tokens for their contributions and for the accuracy and reliability of their AI-driven services or computations.
2. **Smart Contract-driven Payments:** AI services offered within the Endless ecosystem, such as natural language processing, image generation (AIGC), or complex data analysis, are paid for using stablecoins like NUSD or EDS tokens. These payments are handled automatically via smart contracts, ensuring transparency and efficiency. The fees generated from these AI services are distributed to the relevant stakeholders, including the AI model providers, compute node operators, and data providers, based on predefined smart contract rules.
3. **Proof-of-Useful-Work (PoUW):** Endless AI can incorporate a Proof-of-Useful-Work mechanism where computational resources are used for meaningful AI tasks, such as training complex AI models, validating AI outputs, or running resource-intensive inference. Unlike traditional Proof-of-Work which expends energy on arbitrary computations, PoUW in Endless rewards participants for contributing to the advancement and operation of the AI ecosystem, directly replacing traditional mining with valuable work. The validation of these AI computations is performed by the decentralized network, ensuring the integrity and trustworthiness of the results.

3. COMPONENTS AND CORE ARCHITECTURE

Endless AI's robust architecture is built atop the foundational Endless Web3 Genesis Cloud, designed for performance, security, and seamless AI integration. Endless AI's architecture is built around decentralized AI agent swarms—autonomous, collaborative AI entities that operate on the Endless Web3 Genesis Cloud to execute complex tasks without centralized control. These swarms combine multi-agent systems (MAS), blockchain-native governance, and smart contract automation to create a self-coordinating network of AI-driven services. The core components work in concert to support a decentralized, intelligent ecosystem:

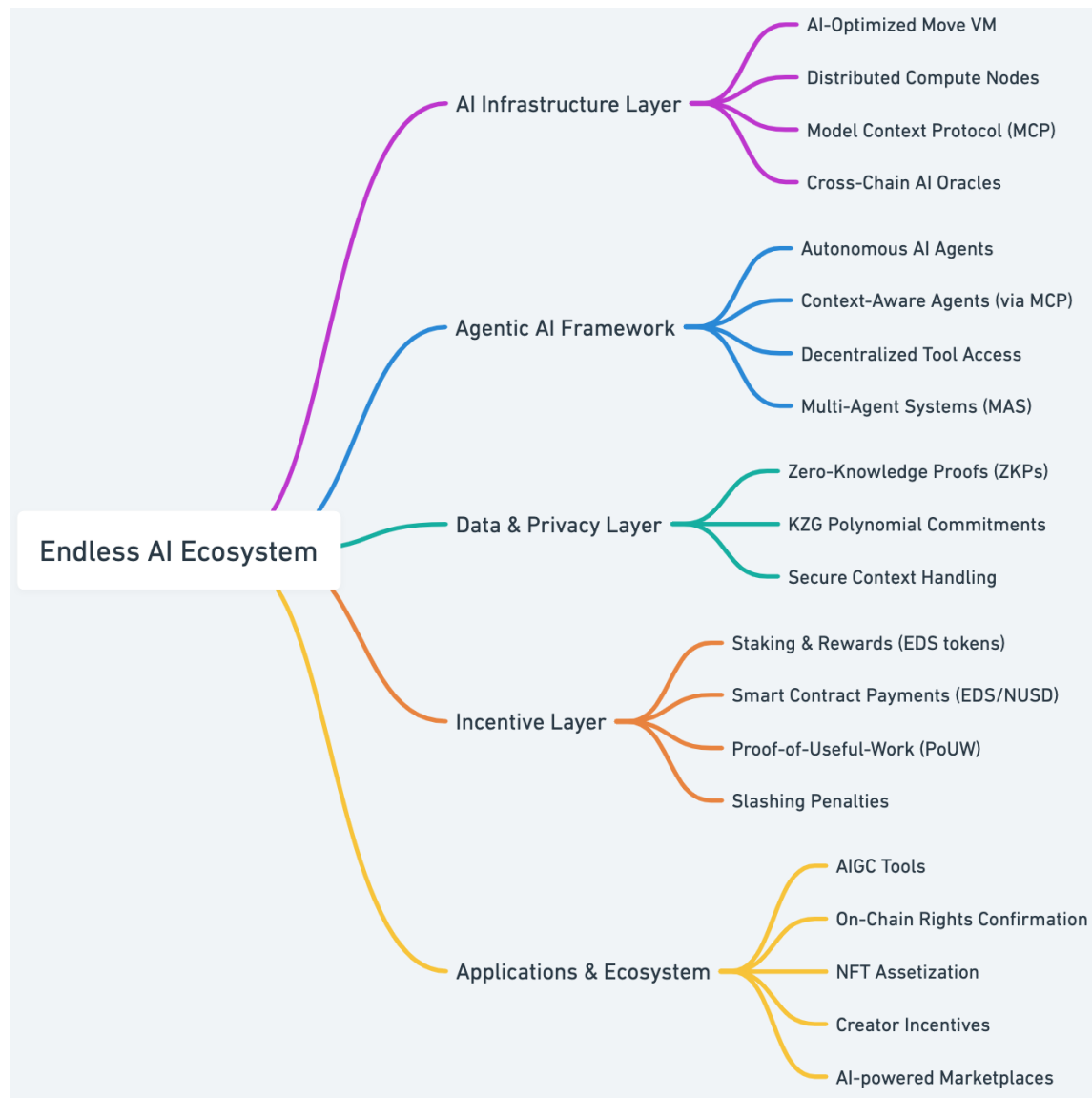


Figure 1: Overview of Endless AI ecosystem

1. AI Infrastructure Layer:

- **AI-OPTIMIZED MOVE VM:** The Endless blockchain utilizes the Move smart contract programming language. The AI-Optimized Move virtual machine is specifically enhanced to improve the execution efficiency of smart contracts that incorporate AI logic or interact with AI models. This optimization is crucial for enabling real-time AI inference and complex AI computations directly within the blockchain environment, facilitating faster and more efficient AI-driven dApps.
- **DISTRIBUTED COMPUTE NODES:** Endless leverages a global network of distributed compute nodes. These nodes contribute their GPU/TPU resources to support the demanding computational requirements of AI tasks, including both training complex AI models and running inference for deployed AI applications. Coordination and verification among these nodes are secured using advanced cryptographic techniques such as *BLS12-381 signatures*, ensuring the integrity and trustworthiness of distributed AI computations.
- **STANDARDIZED INTEGRATION (MODEL CONTEXT PROTOCOL):** The Model Context Protocol is a key enabler for seamless integration of AI within the Endless ecosystem. MCP provides a unified and standardized method for incorporating plugins, tools, and real-world context into AI agents and applications regardless of whether they are running in chat-based dApps, developer environments, or decentralized autonomous systems. This standardization significantly lowers the barrier for developers to integrate AI functionalities and ensures interoperability across different AI applications on Endless.
- **CROSS-CHAIN AI ORACLES:** To enable AI agents to access real-world information and data from outside the Endless blockchain, cross-chain AI oracles are utilized via the Endless Bridge. These oracles securely fetch off-chain data from various sources, such as external APIs, IoT devices, or other blockchain networks, providing AI agents with the necessary context to make informed decisions and generate relevant outputs.

2. Agentic AI Framework:

- **AUTONOMOUS AI AGENTS:** Endless supports the deployment of Autonomous AI Agents often structured as smart contracts. These agents are designed to perceive their environment (on-chain and off-chain data), make independent decisions based on their programming and learned models, and execute actions directly on the blockchain or through connected tools. Use cases include executing trades in DeFi protocols, generating content (AIGC), or managing aspects of Decentralized Autonomous Organi-

zations (DAOs). These agents can also interact with each other via on-chain messaging, enabling complex multi-agent systems.

- **CONTEXT-AWARE AI AGENTS (MCP):** The Model Context Protocol empowers autonomous AI Agents deployed on Endless to be context-aware. MCP allows these agents to access and process real-time external data from diverse sources, such as DeFi protocols, social media feeds, market APIs, and other relevant data streams. This continuous access to up-to-date information significantly enhances the agents' decision-making capabilities and the relevance and accuracy of their outputs.
- **DECENTRALIZED TOOL ACCESS (MCP):** Through the MCP, Endless AI agents gain the ability to invoke and interact with a wide range of external tools and services in a decentralized manner. This includes interacting with decentralized oracles for external data, off-chain computation services for heavy processing, or other specialized protocols, without requiring the tool logic to be hardcoded into the agent's smart contract. This enhances the flexibility, modularity, and capabilities of AI agents on Endless.
- **MULTI-AGENT SYSTEMS:** The Agentic AI framework supports the creation of Multi-Agent Systems where multiple AI agents can collaborate on complex tasks. Consensus among agents on decisions or actions can be achieved using mechanisms like weighted Verifiable Random Functions (wVRF). This enables sophisticated applications like collaborative portfolio rebalancing or decentralized content moderation where multiple AI perspectives contribute to a final outcome.

3. Data & Privacy Layer:

- **ZERO-KNOWLEDGE PROOFS:** ZKPs are a cornerstone of privacy in Endless AI. They allow AI models or agents to prove the correctness of their outputs (e.g., a fraud detection model identifying a suspicious transaction) without revealing the underlying raw data used for the computation or the specifics of the model itself. This is vital for maintaining user privacy and data confidentiality, especially in sensitive applications like healthcare or finance.
- **KATE-ZAVERUCHA-GOLDBERG (KZG) POLYNOMIAL COMMITMENTS:** In Endless Storage Consensus model, KZG polynomial commitments are employed to verify the integrity of AI training data stored on the decentralized network. This cryptographic commitment scheme ensures that the data used to train AI models has not been tampered with, providing a verifiable guarantee of data integrity without revealing the data content.
- **SECURE AND AUDITABLE CONTEXT HANDLING:** In conjunction with Endless's other

privacy-preserving mechanisms like ZKPs and KZG commitments, the Model Context Protocol ensures that interactions where AI agents access and utilize external context remain both verifiable and privacy-compliant. This means that while agents can leverage external data for enhanced performance, the process is auditable on-chain, and user privacy is protected through cryptographic proofs and data isolation.

4. Incentive Layer:

- **TOKENIZED REWARDS:** A robust tokenized reward system is in place. AI agents and developers earn EDS tokens for contributing valuable services to the ecosystem. This includes generating creative content (AIGC), facilitating the creation or management of AI-driven NFTs, assisting in auditing smart contracts, executing trades based on AI analysis, and various other tasks that add value to the Endless platform.
- **SLASHING PENALTIES:** To ensure the reliability and honesty of AI agents and service providers, a slashing mechanism is implemented. Malicious actors or agents that produce incorrect, harmful, or deliberately misleading outputs risk losing a portion of their staked tokens. This provides a strong economic disincentive against dishonest behavior and helps maintain the integrity of the AI ecosystem on Endless.

4. PRODUCTS AND ECOSYSTEM

Endless AI is not merely a technological framework, it is designed to power a vibrant ecosystem of decentralized applications and products that leverage the unique capabilities of integrating AI with Web3. The vision is to foster a self-sustaining ecosystem where innovation is incentivized, data sovereignty is paramount, and creators and users are empowered. This ecosystem is brought to life through various applications built on the Endless Web3 Genesis Cloud with a prime example being Luffa.

4.1. Luffa: A Web3-Native Decentralized SocialFi and Social DID Platform

Luffa stands as a flagship product within the Endless ecosystem embodying the principles of Decentralized Social Finance (SocialFi) and serving as a Social Decentralized Identity (DID) platform. Built on the Endless Web3 infrastructure, Luffa demonstrates how AI, Agentic AI, and decentralized technologies can converge to create a user-centric, privacy-preserving, and economically viable social network. Luffa leverages the core components of Endless AI to deliver its functionalities:

- **Leveraging Agentic AI for SocialFi and User Experience:** Luffa integrates Autonomous AI Agents and potentially Decentralized AI agent swarms to enhance user experience and power its SocialFi features.
 - *Personalized Content Curation:* Agentic AI analyzes user interactions and preferences (while respecting privacy through the Data & Privacy Layer) to curate personalized content feeds, improving user engagement and discovery of relevant content and communities.
 - *Automated SocialFi Interactions:* AI agents can automate aspects of SocialFi activities, such as identifying potential earning opportunities based on user content popularity, managing token rewards for specific social actions (liking, sharing, creating engaging content), or even providing AI-driven insights into social trends and their potential economic value within the platform.
 - *Community Management Assistance:* AI agents can assist in decentralized community management within Luffa by identifying trending discussions, summarizing complex community proposals, or flagging content that may violate community guidelines (based on on-chain rules and potentially decentralized moderation input).
- **Endless AI for AIGC and Creative Expression within Luffa:** As a social platform emphasizing content creation, Luffa directly benefits from Endless AI's support for AIGC and intellectual property management.
 - *Integrated AIGC Tools:* Leveraging the AI Component Marketplace and standardized interfaces via MCP, Luffa can integrate various generative AI models (like LLMs for text or Stable Diffusion for images) directly into the user interface. This empowers users to easily create diverse and engaging content within the platform.
 - *On-Chain Rights Confirmation and Digital Asset Tokenization:* Endless AI components within Luffa facilitate the one-click confirmation of content rights on the blockchain within the creator's workflow. When a user creates content, AI can assist in generating metadata, and the Endless infrastructure handles the on-chain timestamping and linking of the content hash to the user's DID, establishing verifiable proof of creation. This content can then be easily tokenized into NFTs on the Endless blockchain directly within Luffa, leveraging Endless's digital asset tokenization capabilities.
 - *NFT Trading and Circulation:* NFTs representing creative works minted on Endless via Luffa can be seamlessly traded and circulated within the Endless platform's marketplace, creating direct monetization opportunities for creators. The liquidity and discoverability

are enhanced by the underlying Endless infrastructure.

- **Social Decentralized Identity (DID) Powered by Endless’s Data & Privacy Layer:** Luffa functions as a Social DID platform, where user identity and social graph are decentralized and controlled by the user.
 - *Decentralized Data Sovereignty:* User data generated on Luffa (posts, interactions, connections) is stored on Endless’s decentralized storage network, encrypted and controlled by the user’s DID. This aligns with Endless’s principle of Decentralized Data Sovereignty, ensuring users own and manage their social data, unlike traditional social media.
 - *Privacy-Preserving Interactions:* Endless’s Data & Privacy Layer, utilizing ZKPs and secure context handling via MCP, is fundamental to Luffa’s Social DID. AI agents operating on Luffa can analyze user data and behavior for personalization or SocialFi incentives without directly accessing or revealing sensitive raw data, safeguarding user privacy.
 - *Portable Social Graph:* The decentralized nature of the Social DID on Endless means a user’s social graph and reputation built on Luffa could potentially be portable and recognized by other dApps within the Endless ecosystem or across compatible chains, fostering a more interconnected Web3 social experience.
- **Token Incentive System and Community Tokenization:** Luffa utilizes the Endless tokenomics and incentive layer to build a vibrant and economically sustainable social ecosystem.
 - *Rewarding Value Creation:* The platform motivates creators and users to build valuable content and engage meaningfully through the token incentive system. Users earn EDS tokens for creating popular AIGC, receiving engagement (likes, shares), contributing to community moderation, and participating in governance. This tokenization of social activity aligns user incentives with platform growth and value creation.
 - *Community-Driven Value:* The tokenization of the community allows users to have a stake in the platform’s success. Through decentralized governance mechanisms (potentially involving AI agents for analysis and summarization), the community can influence Luffa’s development and operation.

Through Luffa, Endless demonstrates the practical application of its AI and Web3 infrastructure in creating a decentralized social environment that prioritizes user control, privacy, and economic empowerment. Luffa is a key component of the broader Endless ecosystem showcasing how AI-powered decentralized applications can form a truly genuine creative ecosystem where creators are incentivized, intellectual property is protected, and digital assets can be seamlessly monetized and

circulated. The continuous development of Endless AI and its components will directly enhance Luffa's capabilities, leading to more intelligent, personalized, and rewarding social interactions in the Web3 space. Endless AI is central to building a genuine creative ecosystem focused on AIGC, intellectual property rights, on-chain rights confirmation, digital asset tokenization and monetization, and a robust token incentive system for creators.

1. **AIGC with On-Chain Rights:** Endless AI components, particularly generative AI models (like Stable Diffusion integration) and Agentic AI empower creators to generate novel content across various formats (text, images, 3D assets for augmented experiences). The critical innovation lies in enabling creators to confirm the rights of their works directly on the blockchain with minimal effort, potentially with a single click within their familiar development tool workflow. This is facilitated by integrating AI-driven content analysis with blockchain-based timestamping and ownership recording mechanisms. The decentralized and immutable nature of the blockchain, coupled with AI-powered tools for analyzing content originality and tracing provenance, provides a strong foundation for intellectual property protection in the age of AI-generated content.
2. **Digital Asset Token Monetization:** The Endless platform facilitates the assetization of creative works into Non-Fungible Tokens (NFTs) directly on the blockchain. This process can be streamlined and integrated into the creator's workflow through Endless AI components, allowing for easy minting of NFTs representing their original or AI-assisted creations. These digital assets (NFTs) can then be traded, sold, and circulated within the Endless marketplace and potentially across other compatible platforms via cross-chain capabilities. AI agents can potentially assist in pricing, marketing, and managing these digital assets, further enhancing monetization opportunities for creators.
3. **Token Incentive System for Creators:** Endless utilizes its tokenomics (EDS token and NUSD stablecoin) to build a powerful incentive system for creators. Creators are motivated to build valuable content and contribute to the platform through various token-based rewards. This includes earning tokens for creating popular content (AIGC), successfully minting and trading NFTs, receiving tips or payments for their work, and potentially participating in community governance related to content standards or platform development. The tokenization of community engagement ensures that users who contribute to the growth and vibrancy of the creative ecosystem are directly rewarded, fostering a truly genuine and participatory environment. Agentic AI can help in designing and managing these dynamic incentive structures, ensuring fairness and effectiveness.

This integrated approach combining AIGC tools, on-chain rights confirmation, NFT assetization, and a robust token incentive system all powered by Endless AI and the underlying Web3 infrastructure aims to create a self-sustaining and thriving creative ecosystem that empowers creators in unprecedented ways.

5. MARKET AND VALUE TO WEB3 INDUSTRY, DEVELOPERS AND USERS

Endless AI brings significant value to various stakeholders within and beyond the Web3 ecosystem, addressing current market challenges and unlocking new opportunities.

1. For the Web3 Industry:

- **INTEROPERABILITY:** Endless AI acts as a bridge enhancing interoperability not just between different blockchains via the Endless Bridge, but also between different domains within Web3, such as DeFi, NFTs, DAOs, and AIGC. AI-powered agents and applications can seamlessly interact with protocols and assets across these sectors, enabling innovative cross-domain use cases like AI-powered NFT lending platforms or AI agents managing diversified DeFi portfolios linked to digital art markets.
- **TRUSTLESS AI:** By integrating Zero-Knowledge Proofs, Endless AI ensures that the outputs and computations of AI models are verifiable on-chain without revealing sensitive underlying data or model specifics. This introduces a new level of trustlessness to AI applications in Web3, crucial for sensitive operations like fair random number generation (RNG) in gaming or verifiable credit scoring in decentralized lending.
- **BRIDGING WEB2 AND WEB3 WITH INTELLIGENCE:** Endless AI, combined with the broader Endless Web3 Genesis Cloud, significantly lowers the barrier for Web2 applications and developers to transition to Web3. By providing familiar programming language SDKs (Python, JavaScript) and integrating mainstream AI models, Endless makes it easier to build intelligent decentralized applications that can compete with Web2 in terms of user experience and functionality, accelerating mainstream adoption of Web3.

2. For Developers:

- **LOW-CODE/NO-CODE AI TOOLS:** Endless offers SDKs and modular AI components that abstract away much of the underlying complexity of blockchain and AI integration. This provides developers with low-code or even no-code tools to incorporate sophisticated AI functionalities into their dApps, such as integrating ChatGPT-like conversational AI for customer support or leveraging Stable Diffusion for in-app content generation.

- **MONETIZATION OPPORTUNITIES:** Developers can monetize their expertise by creating and offering modular AI components and pre-trained models on the Endless component marketplace. They can earn fees from the sales or usage of their AI components, creating a new revenue stream within the decentralized ecosystem.
- **ENHANCED DEVELOPMENT EFFICIENCY:** The modular architecture, comprehensive SDKs, and AI-powered development tools (potentially in the future, AI-assisted code generation or debugging) on Endless accelerate the development lifecycle for Web3 applications incorporating AI. This allows developers to focus more on innovative application logic rather than wrestling with complex infrastructure.

3. For Users:

- **PERSONALIZED AI EXPERIENCES:** Endless AI enables the creation of personalized AI agents that can cater to individual user needs. These agents can manage DeFi portfolios based on user risk tolerance, curate personalized social feeds based on interests, or even optimize gas fees for transactions, significantly enhancing the user experience and making Web3 more intuitive and efficient.
- **DATA OWNERSHIP AND CONTROL:** Leveraging the decentralized data layer and Self-Sovereign Identity (decentralized identity (DID)) frameworks, Endless AI ensures that users retain ownership and control over their data. AI agents can operate on user-owned data with explicit permissions, and users can potentially monetize their data by providing it for decentralized AI training while maintaining privacy through ZKPs and federated learning.
- **PARTICIPATION IN A CREATIVE ECOSYSTEM:** Users can actively participate in and benefit from the AI-driven creative ecosystem. They can use AIGC tools, own and trade digital assets (NFTs), and be incentivized through token rewards for their contributions to content creation and community building.

6. ROADMAP

The technical roadmap for Endless AI outlines a phased approach to building a sophisticated decentralized intelligent ecosystem deeply integrating AI, Agentic AI, decentralized AI, and the Model Context Protocol with the Endless Web3 infrastructure. This phased implementation focuses on enabling a truly genuine creative ecosystem for creators empowering AIGC, intellectual property protection, on-chain rights confirmation, digital asset token monetization, and a robust

token incentive system. Decentralized AI agent swarms are envisioned as a core component driving many of these advancements.

6.1. Short-Term (1–2 Years)

The initial phase focuses on establishing the foundational elements for AI integration, Web3 frameworks and enabling core AI functionalities within the Endless ecosystem. A key technical focus in this phase is the integration with external Web3 platforms that align with Endless's vision, such as PIXIE. This integration leverages the foundational Endless Web3 infrastructure, AI capabilities, Agentic AI framework, decentralized AI principles, and the Model Context Protocol (MCP) to create synergistic value, particularly for AIGC, intellectual property, and creator monetization within products like Luffa. The integration roadmap with PIXIE involves the following key technical implementations,

- **Seamless AIGC Model and Component Integration:** Integrate PIXIE's in-house AIGC image generation engine as a modular component accessible via the Endless AI Component Marketplace. This involves creating a standardized API or SDK wrapper for the PIXIE AIGC engine that conforms to the MCP. Developers on Endless including those building features for Luffa, can discover and integrate the PIXIE AIGC component from the marketplace. This allows Luffa users, for instance, to generate images using the PIXIE engine directly within the Luffa application interface, leveraging Endless's infrastructure for seamless user experience. Endless AI's AI Component Marketplace facilitates the discovery and access while MCP ensures standardized integration. Future Endless AI agents could potentially interact with the PIXIE AIGC component autonomously for content generation tasks within Luffa or other dApps.
- **Cross-Platform NFT Interoperability and Feature Integration:** Implement technical bridges and protocols to enable interoperability between Endless's NFT standards (Digital Asset standard) and PIXIE's ERC721-based NFT cameras. This includes supporting the display and limited interaction with PIXIE NFT cameras within the Endless ecosystem (e.g., on Luffa user profiles) and allowing Endless-minted NFTs to be displayed or used on PIXIE. The integration will leverage Endless Bridge for cross-chain aspects if PIXIE is on a different chain. Developing smart contracts and middleware to handle asset recognition and metadata exchange between the platforms. For Luffa, this means users could showcase their PIXIE NFT cameras on their profile or within social posts, demonstrating their PIXIE entitlements (Weight, Energy Cap, Speed, Vote). Endless's robust NFT framework provides the base.

Future AI agents on Endless could potentially analyze the social data linked to PIXIE NFTs for valuation purposes or automate aspects of NFT trading on integrated marketplaces.

- **Integration of Social Activity Data and Credit Systems:** Explore secure and privacy-preserving methods for Endless AI agents and applications (including Luffa) to access and interpret social activity data and the Credit System information from PIXIE. This requires defining protocols for data sharing and potentially utilizing Zero-Knowledge Proofs to verify aspects of a user's PIXIE Credit rating or social contributions without revealing sensitive raw data. MCP could facilitate the standardized access to this external context for AI agents. Agentic AI can analyze this data to provide insights or personalize experiences. The Data and Privacy Layer of Endless AI (ZKPs, secure context handling) is crucial for ensuring this integration respects user privacy.
- **Cross-Platform Tokenomics Interaction:** Explore technical mechanisms for interaction between the Endless tokenomics (EDS, NUSD) and the PIXIE tokenomics (PIX). This could involve enabling the use of NUSD or EDS for purchasing certain items or services on PIXIE (e.g., AIGC fees, NFT camera films), or exploring mechanisms for users to earn limited PIX rewards for activities initiated from Endless/Luffa that contribute value to the PIXIE ecosystem. AI agents could potentially optimize strategies for users interacting with the tokenomics of both platforms, such as identifying optimal times to earn or spend tokens based on cross-platform activity and rewards
- **Leveraging AI Agent Swarms for Cross-Platform Activities:** Developing and deploying decentralized AI agent swarms capable of operating across the integrated Endless and PIXIE environments. These swarms would leverage the technical integrations described above (AIGC, NFT, data, tokenomics) to perform complex tasks. This directly utilizes the core concept of decentralized AI agent swarms as a core component of Endless AI, demonstrating their ability to perform complex, value-generating activities across integrated platforms.
- **AI Agent Deployment in Core Products (e.g., Luffa²):** The integration and deployment of basic AI agents within core Endless products like Luffa. In the short term, this could involve AI agents performing tasks like assisting with content tagging, providing basic moderation suggestions based on predefined rules, or offering personalized content recommendations within the Luffa social feed based on user interaction patterns (leveraging the AI-driven content curation concept). The AI Interaction Interface Infrastructure within Luffa will provide an initial entry point for these agents to interact with users and the platform's data (while respecting privacy via E2EE). This phase focuses on demonstrating the practical utility

²<https://luffa.im>

of AI agents within a live, user-facing application on Endless.

This short-term integration roadmap with PIXIE, particularly within the context of products like Luffa, provides concrete technical steps towards realizing the vision of an interconnected, AI-powered creative ecosystem where AIGC, IP rights, tokenization, and incentives are seamlessly integrated across decentralized platforms.

6.2. Mid-Term (3–5 Years)

The mid-term roadmap focuses on building out the decentralized infrastructure for AI training and enabling more complex interactions between autonomous AI agents.

- **AI Component Marketplace:** Establish and launch a decentralized marketplace specifically for AI components. This marketplace will serve as a hub for developers to publish, discover, and integrate pre-trained AI models and modular AI functionalities. This involves developing smart contracts on the Endless chain using the Endless developer kit that underpins Web3 SDK to manage component listings, ownership, licensing, and payment using EDS or NUSD tokens. The marketplace will initially feature popular models such as various large language models (LLMs) and multimodal large models (MLMs) such as Stable Diffusion models provided via standardized SDK interfaces for easy integration into dApps. The Model Context Protocol will be crucial in providing a standardized method for these components to access and utilize context within different dApp environments. This facilitates easy discovery, access, and integration of AI capabilities by developers, lowering the technical barrier for AI-powered dApp development.
- **Deploy Decentralized AI Training Network (PoUW):** Roll out the decentralized AI training network based on a Proof-of-Useful-Work (PoUW) consensus mechanism. This implementation will allow distributed compute nodes across the Endless network to contribute their GPU/TPU resources for training AI models. Participants are incentivized with EDS tokens for providing valuable computational work. This network will support privacy-preserving training techniques like federated learning, where models are trained locally on user-owned data without the raw data leaving the user's device with only model updates or gradients being shared and aggregated. Secure aggregation techniques and potentially ZKPs will be employed to ensure privacy during this process. This phase is critical for enabling the continuous improvement and collective intelligence of AI models within the ecosystem, moving beyond reliance on centrally trained models.

- **Enable AI-to-AI Contracts:** Develop the sophisticated framework and protocols necessary for autonomous AI agents deployed as smart contracts to engage in direct on-chain interactions and negotiations with other AI agents. This involves defining standardized communication protocols between agent smart contracts, leveraging on-chain messaging systems and the MCP for agents to share relevant information and context securely. This capability will pave the way for complex multi-agent systems where AI can autonomously collaborate on tasks, form alliances, or compete based on predefined smart contract rules, enabling more sophisticated decentralized applications.
- **Expansion of Decentralized AI Agent Swarms:** Develop and deploy decentralized AI agent swarms across various Web3 domains on Endless. These swarms as described in the introduction of this section will be implemented as collections of autonomous AI agents designed to collaborate and execute complex strategies.
 - **Decentralized Finance Swarms:** Agents will leverage AI-Optimized Move VM for efficient execution and access real-time data via cross-chain AI oracles. For automated trading and arbitrage, swarms will utilize on-chain data analysis and potentially off-chain computation services for predictive modeling. Yield optimization and liquidity management swarms will constantly interact with various DeFi smart contracts and analyze on-chain data streams. Risk management swarms will employ ZKPs to analyze sensitive transaction patterns without revealing details and integrate with security monitoring tools. Portfolio management swarms will execute transactions directly on-chain based on real-time market data from oracles. Personalized DeFi agents will utilize privacy-preserving techniques like ZKPs to analyze user data with consent.
 - **Web3 Gaming Swarms:** AI agent swarms will power dynamic and intelligent NPCs by executing complex behaviors defined within smart contracts or interacting with on-chain game state. Procedural content generation (PCG) swarms will collaborate, potentially using wVRF for creative randomness and mint generated assets directly as NFTs using Endless's NFT standards. Balanced game economies swarms will analyze on-chain economic data and dynamically adjust parameters encoded in smart contracts. Advanced anti-cheat and security swarms will analyze on-chain player behavior data, potentially leveraging ZKPs for privacy-preserving analysis of sensitive activities. Autonomous guilds and organizations will be managed by AI agents interacting with DAO smart contracts.
 - **Social Applications (DeSoc) Swarms (including Luffa):** Smarter content moderation swarms will analyze content potentially using integrated multimodal AI models (such

as Stable Diffusion) and compare against community-defined rules stored on-chain. Personalized feeds and discovery agents will analyze user interaction data, prioritizing privacy through ZKPs and secure context handling via MCP. Community management and DAO operations swarms will interact with DAO smart contracts to analyze proposals and execute routine tasks. Decentralized reputation swarms will build and maintain reputation systems based on verifiable on-chain interactions. For Luffa, agent swarms will enhance features like personalized content curation enabling more sophisticated AI-powered emojis, KOL and memes to assist in community management tasks, and contribute to the SocialFi token incentive system by identifying valuable user contributions. Interactive social agents will be deployed and potentially monetized on Luffa Superbox and the platform leveraging the AI interaction interface infrastructure.

6.3. Long-Term (5+ Years)

The long-term vision for Endless AI is the realization of a highly advanced, autonomous, and secure decentralized intelligent ecosystem.

- **Agentic AI Super Intelligence:** The long-term vision culminates in the emergence of Agentic AI Super Intelligence within the Endless ecosystem. This represents a state where highly autonomous and intelligent AI agents, leveraging the full capabilities of the Endless AI infrastructure can perform exceptionally complex tasks with minimal human oversight. This involves the maturation of AI-to-AI contracts, sophisticated multi-agent systems, and the ability for agents to learn and adapt continuously through the decentralized training network. These agents will be capable of governing sub-DAOs autonomously, composing and deploying complex cross-chain dApps by orchestrating interactions across different protocols and blockchains via the Endless Bridge, and managing large-scale digital economies. This phase embodies the full potential of decentralized agentic AI.
- **Quantum-Resistant AI Infrastructure:** As quantum computing technology advances, it poses a potential threat to current cryptographic algorithms, including those used in the Endless blockchain (e.g., BLS12-381 signatures for distributed compute node coordination). In the long term, the roadmap includes a significant upgrade of the core cryptographic foundations of Endless AI and the underlying blockchain infrastructure to be quantum-resistant. This involves developing and implementing post-quantum cryptography standards for digital signatures, key exchange, utilising zero-knowledge proofs used within the AI ecosystem. This proactive technical measure is essential to ensure the long-term security,

integrity, and resilience of the decentralized AI network and all applications built on Endless against the threat of quantum attacks.

- **Fully Autonomous Creative Ecosystem:** In the long term, the advancements in AI, Agentic AI and decentralized infrastructure will fully realize the vision of a truly genuine creative ecosystem. AI agents will seamlessly assist creators throughout the AIGC process, from ideation to generation and refinement. On-chain rights confirmation will be an automated and integral part of the creation workflow, potentially utilizing AI to analyze content originality and embedding verifiable proofs directly into the digital assets (NFTs). Digital asset tokenization and monetization will be highly efficient and potentially managed in part by AI agents optimizing discoverability and pricing in the marketplace. The token incentive system will be dynamically adjusted by AI agents to ensure fair and effective rewards for creators and community contributors. Agent swarms within our products such as Luffa will facilitate highly personalized and engaging social interactions centered around creative content, further fueling the ecosystem's growth and value creation.